

Extended Data 5 | Marker-validity test: adjacent-minus-distant signal and permutation p (admissible if p < 0.05 over 1,000 permutations)

breast: 28/38 admissible (rejected shaded)

gene	signal	p	adm
AQP1	+0.059	0.001	Y
AVPR1A	+0.213	0.001	Y
BTNL9	+0.009	0.001	Y
CAVIN2	+0.049	0.001	Y
CD79A	+0.087	0.001	Y
CD93	+0.060	0.001	Y
CLEC14A	+0.045	0.001	Y
DERL3	+0.030	0.001	Y
ESM1	+0.070	0.001	Y
HOXD9	+0.032	0.001	Y
ITM2C	+0.074	0.001	Y
KDR	+0.060	0.001	Y
KRT14	+0.217	0.001	Y
KRT15	+0.261	0.001	Y
KRT23	+0.195	0.001	Y
KRT5	+0.197	0.001	Y
MMRN2	+0.040	0.001	Y
MYH11	+0.176	0.001	Y
NOSTRIN	+0.027	0.001	Y
OPRPN	+0.295	0.001	Y
PIGR	+0.109	0.001	Y
RAMP2	+0.059	0.001	Y
SLAMF7	+0.070	0.001	Y
SOX17	+0.035	0.001	Y
SOX18	+0.025	0.001	Y
TNFRSF17	+0.071	0.001	Y
VWF	+0.050	0.001	Y
CLDN5	+0.014	0.008	Y
MZB1	+0.002	0.104	n
UCP1	+0.314	0.152	n
CRHBP	+0.264	0.242	n
SLC4A1	+0.012	0.454	n
MPO	-0.058	0.531	n
AHSP	-0.091	0.599	n
EGFL7	-0.001	0.642	n
CYP1A1	-0.373	0.996	n
MUC6	-0.570	0.996	n
TENT5C	-0.046	1.000	n

lung: 31/32 admissible

gene	signal	p	adm
ACKR1	+0.068	0.001	Y
BANK1	+0.098	0.001	Y
BMX	+0.107	0.001	Y
BTNL9	+0.060	0.001	Y
CD19	+0.066	0.001	Y
CD79A	+0.141	0.001	Y
CD8A	+0.069	0.001	Y
CNN1	+0.117	0.001	Y
CTLA4	+0.055	0.001	Y
DERL3	+0.023	0.001	Y
DES	+0.037	0.001	Y
FBLN1	+0.132	0.001	Y
FOXP3	+0.067	0.001	Y
GZMK	+0.138	0.001	Y
IL7R	+0.120	0.001	Y
LGI4	+0.096	0.001	Y
LILRA4	+0.061	0.001	Y
MFAP5	+0.091	0.001	Y
MMRN1	+0.068	0.001	Y
MS4A1	+0.096	0.001	Y
MYH11	+0.080	0.001	Y
MZB1	+0.085	0.001	Y
PCSK2	+0.065	0.001	Y
SFRP2	+0.097	0.001	Y
SLAMF7	+0.077	0.001	Y
SPIB	+0.262	0.001	Y
TENT5C	+0.023	0.001	Y
TNFRSF13B	+0.077	0.001	Y
TNFRSF17	+0.067	0.001	Y
CYP4B1	+0.033	0.002	Y
DPEP1	+0.084	0.002	Y
TAC1	-0.025	0.906	n